

Requirements Document

Project: EasyStego: A Drag-and-Drop Image Steganography Tool

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1. Project Overview

EasyStego is a Java-based desktop application that allows users to hide and retrieve secret text messages inside PNG image files using the Least Significant Bit (LSB) steganography technique. It provides an intuitive drag-and-drop GUI built with Java Swing, requiring no prior technical knowledge from the user.

2. Functional Requirements

Functional requirements define the specific behaviours and features the system must support.

ID	Requirement Description	Category	Priority
FR-01	User can drag and drop a PNG or JPEG image into the Hide Message tab	Input	High
FR-02	Application displays the maximum embeddable message capacity in real time after image load	Display	High
FR-03	User can type a secret text message into the message text area	Input	High
FR-04	Application embeds the message into the image using the LSB algorithm	Core Logic	High
FR-05	Application appends the ####END#### terminator to the message before embedding	Core Logic	High
FR-06	Stego-image is saved as a lossless PNG file via a file save dialog	Output	High
FR-07	User can drag and drop a stego-image into the Reveal Message tab	Input	High

ID	Requirement Description	Category	Priority
FR-08	Application auto-decodes the hidden message immediately on image drop	Core Logic	High
FR-09	Decoded message is displayed in a read-only text area	Display	High
FR-10	User can copy the decoded message to clipboard with one click	Utility	Medium
FR-11	Application shows an error dialog if no image is loaded before embedding	Validation	High
FR-12	Application shows an error dialog if the message text area is empty	Validation	High
FR-13	Application shows an error or fallback message if no hidden data is found	Validation	Medium
FR-14	Image preview is displayed inside the drop zone after a file is loaded	Display	Medium

3. Non-Functional Requirements

Non-functional requirements define the quality attributes and constraints of the system.

ID	Requirement Description	Category	Priority
NFR-01	Output stego-image must be visually identical to the original image	Quality	High
NFR-02	Application must run on Windows, macOS, and Linux with JDK 11+	Compatibility	High
NFR-03	Output file format must be PNG (lossless) to preserve embedded bits	Reliability	High
NFR-04	Embedding and extraction must complete without data loss for valid inputs	Reliability	High
NFR-05	The GUI must be operable without any prior technical knowledge	Usability	High

ID	Requirement Description	Category	Priority
NFR-06	The application must launch as a single executable JAR file	Portability	Medium
NFR-07	Capacity calculation must update in real time after each image load	Performance	Medium
NFR-08	JPEG input images must be converted to PNG on save to prevent data loss	Reliability	Medium
NFR-09	Source code must be version-controlled using Git and hosted on GitHub	Maintainability	Medium

4. Constraints

- Language: Java (JDK 11 or higher) only.
- GUI Framework: Java Swing (javax.swing) — no third-party UI libraries.
- Image Format: Only PNG images support lossless bit embedding; JPEG inputs are converted on save.
- Data Type: Only plain text messages are supported; binary file embedding is out of scope.
- Encryption: No encryption is applied to the message before embedding in this version.

5. Assumptions

- The user has Java JDK 11 or higher installed on their machine.
- Input images are assumed to be in PNG or JPEG format; other formats may not load correctly.
- The message length is assumed not to exceed the image capacity (width x height / 8 characters).
- The stego-image must not be re-saved as JPEG after embedding, as lossy compression will destroy the hidden data.

6. Out of Scope

- Binary file embedding (images, PDFs, audio).
- AES or any other form of message encryption.
- Password-protection for message extraction.
- Multi-bit LSB embedding (2-bit or 3-bit per channel).
- Mobile or web-based deployment.